The Cato Institute's Overlapping Generations Model for Social Security Reform

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1 Introduction

The aim of this model is to provide the public with a computable overlapping generations (OLG) model that is capable of scoring most Social Security reforms without using too much computing power. The OLG model we have built here takes most of its structure from the Penn-Wharton Budget Model's (PWBM) OLG. However, we make several important changes:

- Unlike the PWBM's OLG, our OLG allows for the analysis of policies that change dependent and survivor benefits.
- Moreover, our OLG connects explicitly with Social Security Trust Fund rates, and explicitly models the taxation of benefits.
- Unlike the PWBM model, we do not solve the full transition path due to computing limitations. Instead, we use a back-of-the-envelope methodology from Mankiw and Weinzierl (2006) to approximate transition dynamics between steady states.

2 Production

Our production function is a simple Cobb-Douglas with no technology:

$$Y = K^{\alpha} L^{1-\alpha}$$

where K is aggregate capital, L is aggregate labor in efficiency units, and $\alpha = 0.36$ is the capital share. Firms are competitive and hire factors up to their marginal products:

$$w = (1-\alpha)\left(\frac{K}{L}\right)^{\alpha}, \qquad r_K = \alpha\left(\frac{L}{K}\right)^{1-\alpha}$$

2.1 Income Decomposition

Households hold a single asset a_j. The composition of the economy's asset pool determines how each household's savings map into different income streams. Define the portfolio shares:

$$\psi_{cap} = \frac{K}{K+D}, \qquad \psi_{debt} = \frac{D}{K+D}$$

where D is government debt. A fraction ψ_{cap} of each household's assets is implicitly invested in productive capital, and ψ_{debt} in government bonds. There is no portfolio choice—all households hold the same mix.

The capital share is further split between the corporate sector ($\zeta_{corp} = 0.55$) and the pass-through sector ($\zeta_{pass} = 0.45$). Both sectors earn the net-of-depreciation return $\pi = r_K - \delta$, but face different tax treatment. Combined with labor income $y^{lab}_{k,j} = w \cdot z_{k,j} \cdot n_{k,j}$ for spouse k at age j, this yields four income flows:

$$y^{lab}_{k,j} = w \cdot z_{k,j} \cdot n_{k,j} \tag{1}$$

$$y^{corp}_{j} = \pi \cdot \zeta_{corp} \cdot \psi_{cap} \cdot a_j \tag{2}$$

$$y^{pass}_{j} = \pi \cdot \zeta_{pass} \cdot \psi_{cap} \cdot a_j \tag{3}$$

$$y^{debt}_{j} = r_G \cdot \psi_{debt} \cdot a_j \tag{4}$$

where $r_G = 0.03$ is the exogenous government bond rate and $\delta = 0.06$ is the depreciation rate.

2.2 Government Sector

The government receives revenue from income, payroll, and corporate taxes. It spends on Social Security outlays and residual government consumption (fixed at 18% of GDP). Like the closed version of the Penn-Wharton Budget Model, the government balances its budget through raising extra revenues, the source of which is undetermined. Government debt evolves as:

$$D_{t+1} = D_t(1 + r^{nom}_t) + G_t + S_t - T_t$$

where T_t is total tax revenue, S_t is Social Security outlays, $G_t = 0.18 \cdot Y_t$, and $r^{nom}_t = (1 + r_G)(1 + g_{\pi,t}) - 1$.

3 Taxation

The tax function τ_j in this model mirrors the structure of the U.S. federal tax code. Total taxes paid by a household at age j consist of four components:

$$\tau_j = T^{ord}_j + T^{pref}_j + T^{payroll}_j + T^{con}_j$$

3.1 Income Classification

Each income source is assigned an *inclusion rate* that determines what fraction enters the taxable base. These ratios approximates the standard deduction. Corporate equity income is split between ordinary income (dividends taxed at ordinary rates) and preferential income (long-term capital gains and qualified dividends). :

Income Source	Ordinary Inclusion (θ^{ORD})	Preferential Inclusion (θ^{PREF})
Labor income	0.95	—
Corporate equity	0.60	0.40
Pass-through	1.00	—
Bond interest	1.00	—
Social Security	Formula	—

3.2 Social Security Benefit Taxation

The taxable portion of Social Security benefits is determined by the modified adjusted gross income (MAGI) test:

$$\text{MAGI}_j = y_j^{other} + 0.5 \cdot b_j^{SS}$$

where y_j^{other} is all non-Social-Security ordinary income and b_j^{SS} is total Social Security benefits. The taxable amount is:

$$b_j^{SS,tax} = \begin{cases} 0 & \text{if MAGI}_j \leq \$32{,}000 \\ \min\left(0.50 \cdot b_j^{SS},\ 0.50 \cdot (\text{MAGI}_j - \$32{,}000)\right) & \text{if } \$32{,}000 < \text{MAGI}_j \leq \$44{,}000 \\ \min\left(b_j^{SS,50\%} + 0.85 \cdot (\text{MAGI}_j - \$44{,}000),\ 0.85 \cdot b_j^{SS}\right) & \text{if MAGI}_j > \$44{,}000 \end{cases}$$

where $b_j^{SS,50\%}$ denotes the tier-1 taxable amount. The thresholds \$32,000 and \$44,000 are the married-filing-jointly thresholds. Critically, these thresholds are *not indexed* to inflation or wage growth.

The taxable amount is further decomposed into an OASI component (the first 50% tier) and an HI component (the remainder up to the 85% tier), since the resulting tax revenues are credited to different trust funds.

3.3 Ordinary Income Tax

Total ordinary income at for household at time j is:

$$y_j^{ord} = \theta_{lab}^{ORD} \cdot (y_{1,j}^{lab} + y_{2,j}^{lab}) + \theta_{corp}^{ORD} \cdot y_j^{corp} + \theta_{pass}^{ORD} \cdot y_j^{pass} + y_j^{debt} + b_j^{SS,tax}$$

This is taxed through a progressive bracket schedule. For married-filing-jointly households:

Bracket Floor	Marginal Rate
\$0	10%
\$19,400	12%
\$78,950	22%
\$168,400	24%
\$321,450	32%
\$408,200	35%
\$612,350	37%

Survivors (widows/widowers) file as single filers and face narrower brackets with lower thresholds. The ordinary income tax is computed as:

$$T_j^{ord} = \sum_i r_i \cdot \max\left(0,\ \min(y_j^{ord},\ b_{i+1}) - b_i\right)$$

where r_i is the marginal rate and b_i is the floor of bracket i.

3.4 Preferential Income Tax

The preferential portion of corporate equity income ($\theta_{corp}^{PREF} \cdot y_j^{corp}$) is taxed at lower rates:

Bracket Floor (MFJ)	Rate
\$0	0%
\$78,750	15%
\$488,850	20%

3.5 Payroll Tax

Each spouse's labor income is subject to a payroll tax up to the taxable maximum:

$$T_j^{payroll} = \tau_p \cdot \min(y_{1,j}^{lab},\ \overline{y}) + \tau_p \cdot \min(y_{2,j}^{lab},\ \overline{y})$$

where $\tau_p = 0.106$ is the combined employer-employee OASI rate and $\overline{y} =$ \$176,100 is the taxable earnings cap. The cap is indexed to average wage growth in the projection. Note that this rate covers only the OASI portion of FICA; the HI (Medicare) component is handled separately through the benefit taxation channel.

3.6 Corporate Income Tax

On the production side, corporate profits are taxed at a statutory rate of $\tau_c = 0.21$. This tax does not enter the household optimization directly but affects the general equilibrium through the cost of capital and the government budget constraint.

4 Social Security

4.1 Average Indexed Monthly Earnings

Social Security benefits are computed from each worker's Average Indexed Monthly Earnings (AIME). For each spouse k, the model records annual earnings $e_{k,j} = w \cdot z_{k,j} \cdot n_{k,j}$ over the working life, subject to the taxable maximum. Earnings prior to age 60 are indexed to economy-wide wage growth:

$$\tilde{e}_{k,j} = \begin{cases} \min\left(e_{k,j} \cdot (1+g_w)^{60-j},\ \overline{y}\right) & \text{if } j < 60 \\ \min\left(e_{k,j},\ \overline{y}\right) & \text{if } j \geq 60 \end{cases}$$

The AIME is then the average of the highest 35 years of indexed earnings:

$$\text{AIME}_k = \frac{1}{35} \sum_{i=1}^{35} \tilde{e}_{k,(i)}$$

where $\tilde{e}_{k,(i)}$ denotes the i-th highest indexed earning year.

4.2 Primary Insurance Amount

The Primary Insurance Amount (PIA) is computed from AIME using the standard bend-point formula:

$$\text{PIA}_k = 0.90 \cdot \min(\text{AIME}_k, b_1) + 0.32 \cdot \max(0, \min(\text{AIME}_k, b_2) - b_1) + 0.15 \cdot \max(0, \text{AIME}_k - b_2)$$

where $b_1 = \$14{,}712$ and $b_2 = \$88{,}692$ are the annualized bend points in 2025.

5 Household Problem and Solution

5.1 Preferences

Each person has the following flow utility:

$$u(c_j, n_{k,j}) = \frac{c_j^{1-\sigma} - 1}{1-\sigma} + \gamma \ln(1 - n_{k,j})$$

where c_j denotes consumption at age j, $n_{k,j} \in [0, 1)$ denotes hours worked by spouse k, $\sigma = 2$ is the coefficient of relative risk aversion, and $\gamma = 0.30$ governs the taste for leisure.

The representative household consists of a married couple. Both spouses consume a single household good c_j and independently supply labor. The couple maximizes a unitary household utility function:

$$U(c_j, n_{1,j}, n_{2,j}) = \frac{c_j^{1-\sigma} - 1}{1-\sigma} + \gamma \left[\ln(1 - n_{1,j}) + \ln(1 - n_{2,j})\right]$$

Like the Penn-Wharton model, households enter the economy at age 21, retire at 67, and live to a maximum age of 100. The discount factor is $\beta = 0.97$.

5.2 Productivity

Individual productivity for spouse k at age j is:

$$z_{k,j} = \exp(\epsilon_j + \bar{z}_k + \tilde{z}_{k,j})$$

where ϵ_j is a deterministic age-efficiency profile (quadratic in normalized age), $\bar{z}_k$ is a permanent type drawn at birth, and $\tilde{z}_{k,j}$ is a persistent shock following:

$$\tilde{z}_{k,j} = \rho\, \tilde{z}_{k,j-1} + \eta_{k,j}, \qquad \eta_{k,j} \sim N(0, \sigma_\eta^2)$$

with $\rho = 0.990$ and $\sigma_\eta^2 = 0.007$. For computational ease, we discretize this process onto a grid of $n_z = 5$ points using Tauchen's method, which produces a transition matrix Π with entries $\Pi_{z,z'} = \Pr(\tilde{z}_{k,j+1} = z' \mid \tilde{z}_{k,j} = z)$.

The permanent type $\bar{z}_k$ is drawn from a discrete distribution with four types:

Type	Low	Medium	High	Very High
$\bar{z}_k$	-0.45	0.10	0.85	1.20
Probability	0.40	0.45	0.12	0.03

All households are couples. There are $4 \times 4 = 16$ couple types, formed by pairing permanent productivity types $(\bar{z}_1, \bar{z}_2)$. The joint probability of a couple with types (i, j) incorporates assortative mating:

$$P(i,j) \propto p_i \cdot p_j \cdot \exp\left(\rho_a \cdot \left(1 - \frac{|i-j|}{n-1}\right)\right)$$

where $\rho_a = 0.6$ and $n = 4$. There is no marriage market, divorce, or remarriage; couples dissolve only through spousal death.

5.3 Labor Supply

Once productivity has been determined, wages follow from the marginal product of labor. Labor supply is determined by the intratemporal first-order condition:

$$n_{k,j} = 1 - \frac{\gamma}{w\, z_{k,j}\,(1 - MTR_{k,j}) \cdot c_j^{-\sigma}}$$

This equates the marginal disutility of labor, $\gamma/(1 - n_{k,j})$, to the after-tax marginal return to working, $w\, z_{k,j}(1 - MTR_{k,j})$, valued at the marginal utility of consumption $c_j^{-\sigma}$. Because $MTR_{k,j}$ depends on labor income, which depends on $n_{k,j}$, this is solved by fixed-point iteration.

5.4 Budget Constraint

A couple at age j with assets a_j faces:

$$c_j + a_{j+1} + \tau_j = w\, z_{1,j}\, n_{1,j} + w\, z_{2,j}\, n_{2,j} + (1+\pi)a_j + b_j^{SS}$$

$$a_{j+1} \geq 0$$

where $\tau_j = \tau(w\, z_{1,j}\, n_{1,j},\ w\, z_{2,j}\, n_{2,j},\ b_j^{SS},\ c_j)$ is the tax function described in Section 3 and b_j^{SS} denotes Social Security benefits (zero during working years, positive during retirement).

5.5 Bellman Equation

The couple's problem is:

$$V^c(a_j, z_{1,j}, z_{2,j}, j) = \max_{c_j,\, n_{1,j},\, n_{2,j},\, a_{j+1}} U(c_j, n_{1,j}, n_{2,j}) + \beta\, \mathbb{E}_j[V_{j+1}]$$

subject to the budget constraint above. The continuation value reflects gender-specific mortality. At each age j, either both spouses survive, only spouse 1 survives, or only spouse 2 survives:

$$\mathbb{E}_j[V_{j+1}] = s_j^m\, s_j^f\, \mathbb{E}\big[V_{j+1}^c\big] + s_j^m(1 - s_j^f)\, \mathbb{E}\big[V_{j+1}^{s_1}\big] + (1 - s_j^m)s_j^f\, \mathbb{E}\big[V_{j+1}^{s_2}\big]$$

where s_j^m and s_j^f are male and female survival probabilities from the SSA's 2022 period life tables.

5.6 Endogenous Grid Method (EGM)

We solve the household problem by backward induction from age $J = 100$. At the terminal age, households consume all available resources. Working backwards, we employ the endogenous grid method (EGM) to avoid root-finding.

EGM relies on analytically inverting the Euler equation. The intertemporal first-order condition is:

$$\begin{aligned} c_j^{-\sigma} = \beta(1+\pi)\Big[& s_j^m s_j^f \sum_{z_1', z_2'} \Pi_{z_1, z_1'} \Pi_{z_2, z_2'}\, (c_{j+1}^c)^{-\sigma} \\ & + s_j^m (1 - s_j^f) \sum_{z_1'} \Pi_{z_1, z_1'}\, (c_{j+1}^{s_1})^{-\sigma} \\ & + (1 - s_j^m) s_j^f \sum_{z_2'} \Pi_{z_2, z_2'}\, (c_{j+1}^{s_2})^{-\sigma} \Big] \end{aligned} \tag{5}$$

where c_{j+1}^c, $c_{j+1}^{s_1}$, and $c_{j+1}^{s_2}$ denote next-period consumption under the couple and widow scenarios, respectively. The two spouses' persistent productivity shocks are independent, so the joint transition probability is $\Pi_{z_1, z_1'} \cdot \Pi_{z_2, z_2'}$.

For each candidate a_{j+1} on the exogenous asset grid, we compute c_j from the Euler equation, solve for optimal hours $(n_{1,j}^*, n_{2,j}^*)$ by iterating on the intratemporal conditions, and back out today's assets from the budget constraint:

$$a_j = \frac{a_{j+1} + c_j + \tau_j - w\, z_{1,j}\, n_{1,j} - w\, z_{2,j}\, n_{2,j} - b_j^{SS}}{1 + \pi}$$

This produces an endogenous grid $\{a_i^{endo}, c_i, n_{1,i}, n_{2,i}\}$, which is interpolated back onto the original asset grid. For households at the borrowing constraint ($a_{j+1} = 0$), we solve directly by iterating on the budget constraint.

The survivor problems are solved first, as their value functions enter the couple's continuation value. Because Social Security benefits depend on lifetime earnings, which depend on labor supply, the model uses a two-pass procedure: first solving with fixed hours ($n = 0.35$) to compute initial AIMEs, then resolving with equilibrium hours and updated benefits.

6 Steady State and General Equilibrium

We find the steady state by iterating on the capital-labor ratio K/L:

1. **Initialize.** Guess an initial capital-labor ratio $(K/L)^0$.

2. **Prices.** Given $(K/L)^i$, compute factor prices:

$$w^i = (1-\alpha)\left(\frac{K}{L}\right)_i^{\alpha}, \qquad r_K^i = \alpha\left(\frac{L}{K}\right)_i^{1-\alpha}, \qquad \pi^i = r_K^i - \delta$$

   Compute $\psi_{cap} = K/(K+D)$ and $\psi_{debt} = D/(K+D)$, where $D = 0.78 \cdot Y$.

3. **Household problems.** For each of the 16 couple types $(\bar{z}_1, \bar{z}_2)$, taking prices as given:

   (a) Solve the two survivor problems by EGM backward induction.

   (b) Solve the couple problem by EGM, using the survivor value functions in the continuation value.

   (c) Compute AIME via the two-pass procedure (fixed hours, then equilibrium hours).

   (d) Re-solve with updated benefits.

   (e) Compute the stationary distribution over $(a_j, z_{1,j}, z_{2,j}, j, m)$ by forward iteration, where $m \in \{\text{couple}, \text{survivor}_1, \text{survivor}_2\}$.

   The 16 couple-type problems are solved in parallel.

4. **Aggregation.** Aggregate across couple types, weighting by $P(i,j)$:

$$A^i = \sum_{\text{types}} P(i,j) \sum_{j,m} \mu(a_j, z_{1,j}, z_{2,j}, j, m) \cdot a_j$$

$$L^i = \sum_{\text{types}} P(i,j) \sum_{j,m} \mu(a_j, z_{1,j}, z_{2,j}, j, m) \cdot (z_{1,j}\, n_{1,j} + z_{2,j}\, n_{2,j})$$

5. **Implied capital-labor ratio.** $K^{implied} = \psi_{cap} \cdot A^i$, and $(K/L)^{implied} = K^{implied}/L^i$.

6. **Calibrate μ.** Update $\mu = \$63{,}000 \,/\, \bar{e}_{model}$ so that model average earnings match the U.S. average.

7. **Update.** $(K/L)^{i+1} = 0.7 \cdot (K/L)^i + 0.3 \cdot (K/L)^{implied}$.

8. **Convergence.** Repeat until $|(K/L)^{implied} - (K/L)^i| \,/\, |(K/L)^i| < 10^{-4}$.

6.1 Stationary Distribution

The distribution over $(a_j, z_{1,j}, z_{2,j}, j, m)$ is computed by forward iteration from age 21 to 100. At age 21, all mass enters as couples, uniform over $(z_{1,j}, z_{2,j})$. At each subsequent age:

- **Productivity:** Each spouse's persistent shock transitions via Π. For couples, the joint transition is $\Pi_{z_1,z_1'} \cdot \Pi_{z_2,z_2'}$.
- **Assets:** Mass flows to $a'(a_j, z_{1,j}, z_{2,j}, j)$, distributed to adjacent grid points via linear interpolation.
- **Mortality:** With probability $s_j^m s_j^f$, both survive (couple). With probability $s_j^m(1 - s_j^f)$, mass flows to survivor$_1$. With probability $(1 - s_j^m)s_j^f$, mass flows to survivor$_2$.
- **Population growth:** Each cohort is discounted by $1/(1 + g_n)$.

Distributions are normalized to sum to one.

7 Projection Methodology

The projection engine takes the steady-state solution and extrapolates it forward over a 75-year horizon (2025–2099).

7.1 Exogenous Growth Paths

Three exogenous growth rates, derived from the 2026 Social Security Trustees' Report intermediate scenario, drive the projection:

Variable	Symbol	Baseline Value
Productivity growth	g_A	1.12%
Population growth	g_n	0.5%
Inflation (CPI)	g_π	2.3%

Cumulative indices are:

$$A_t = \prod_{\tau=1}^{t}(1 + g_{A,\tau}) \qquad \text{(cumulative productivity)} \tag{6}$$

$$P_t = \prod_{\tau=1}^{t}(1 + g_{n,\tau}) \qquad \text{(cumulative population)} \tag{7}$$

$$\Pi_t = \prod_{\tau=1}^{t}(1 + g_{\pi,\tau}) \qquad \text{(cumulative price level)} \tag{8}$$

$$W_t = \prod_{\tau=1}^{t}(1 + g_{A,\tau} + g_{\pi,\tau}) \qquad \text{(cumulative nominal wage)} \tag{9}$$

7.2 Demographics

The dependency ratio (retirees per worker) evolves along a logistic path:

$$d_t = d_0 + \frac{d_\infty - d_0}{1 + \exp(-\kappa(t - \bar{t}))}$$

where $d_0 \approx 0.29$, $d_\infty = 0.45$, $\kappa = 0.08$, and $\bar{t} = 25$. The number of retirees scales as $N_t^{ret} = N_{ss}^{ret} \cdot (d_t/d_0) \cdot N_{hh,t}$.

7.3 Trust Fund Dynamics

Cash flow is $CF_t = R_t^{FICA} + R_{btax,t}^{OASI} - S_t$. Interest accrues only while $TF_t > 0$:

$$TF_{t+1} = TF_t + CF_t + r_{TF} \cdot \max(TF_t, 0)$$

The depletion year is the first t such that $TF_t < 0$.

7.4 Reform Scoring

Unlike most other models, we do not solve for the transition path directly. Instead, we adopt the back-of-the-envelope approach from Mankiw and Weinzierl (2006). We solve two steady states—one under current law (K^{base}, L^{base}) and one under the reform (K^{ref}, L^{ref})—and assume that capital and labor converge to the reform steady state at rate λ:

$$\ln K_t - \ln K^{ref} = \left(\ln K^{base} - \ln K^{ref}\right) e^{\lambda t}$$

$$\ln L_t - \ln L^{ref} = \left(\ln L^{base} - \ln L^{ref}\right) e^{\lambda t}$$

where $\lambda < 0$ is the negative eigenvalue of the Jacobian of the linearized Ramsey system. The dynamic system in capital and consumption is:

$$\dot{k} = k^\alpha - c - (\delta + g_A)k$$

$$\dot{c} = \frac{c}{\sigma}\left[(1 - s_k)\alpha k^{\alpha-1} - \delta - \rho - \sigma g_A\right]$$

where $\rho = 1/\beta - 1$. The Jacobian evaluated at the steady state is:

$$\mathbf{J} = \begin{pmatrix} \rho_{eff} & -1 \\ \frac{c^*}{\sigma}(1 - s_k)\alpha(\alpha - 1)(k^*)^{\alpha-2} & 0 \end{pmatrix}$$

where $\rho_{eff} = \rho + (\sigma - 1)g_A$. The negative eigenvalue is:

$$\lambda = \frac{\rho_{eff} - \sqrt{\rho_{eff}^2 + 4\frac{c^*}{\sigma k^*}(1 - \alpha)r_K^*}}{2}$$

Given the transition paths of K_t and L_t, all macro aggregates are derived from the production function rather than by blending steady-state ratios:

$$Y_t = K_t^{\alpha} L_t^{1-\alpha}, \qquad w_t = (1-\alpha)(K_t/L_t)^{\alpha}, \qquad r_{K,t} = \alpha(L_t/K_t)^{1-\alpha}$$

Payroll is $w_t \cdot L_t$, corporate profits follow from $Y_t - w_t L_t - \delta K_t$, and fiscal variables are computed from these aggregates. Distributional variables (payroll tax rate, fraction of earnings above the cap, benefit levels per retiree, benefit taxation rates) are interpolated between the two steady states using the weight $\omega_t = 1 - e^{\lambda(t-t^{ref})}$.

7.5 Output Metrics

The projection produces year-by-year series for: nominal GDP, average earnings, the payroll tax cap, taxable payroll, FICA revenue, SS outlays, average benefit per retiree, benefit tax revenue (OASI and HI), trust fund interest, cash flow, trust fund balance, effective FICA rate, cost rate, debt, and debt-to-GDP. Summary statistics are reported at 10-year, 30-year, and 75-year actuarial windows.